PROJECT SUMMARY & BUSINESS CONTEXT

A structured overview of the BMW Sales Performance & Market Analysis project

Project period: 2010–2024 **Category:** Business Intelligence & Data Analytics

Tools: MySQL, Tableau Public, Microsoft Excel **Source format:** CSV

1. Executive Summary

This project analyzes a BMW vehicle-sales dataset covering the period 2010–2024. The goal is to organize available sales and vehicle attributes into a repeatable analysis workflow and present the results through SQL queries and interactive Tableau dashboards. The work is structured around overall performance, model-level comparisons, regional distribution, and vehicle attributes.

The deliverable combines query-based analysis with dashboard-based exploration. SQL is used to prepare and inspect analytical outputs, while Tableau presents selected metrics and comparisons in a visual format. The report intentionally distinguishes analytical capability from verified business conclusions: specific winners, growth rates, and recommendations should only be stated after checking the actual query results and underlying data.

2. Business Problem and Rationale

A vehicle-sales dataset can contain many records and dimensions, making it difficult to compare performance by year, model, region, fuel type, transmission, color, engine size, mileage, and price using raw rows alone. A structured analytics project helps turn the dataset into consistent summaries that can be reviewed by business users.

- Create a consolidated view of the available sales measures.
- Compare model and regional performance using consistent aggregation logic.
- Explore how available vehicle attributes vary across the dataset.
- Provide interactive dashboards so viewers can filter and inspect patterns.
- Keep SQL queries, dashboard exports, and documentation organized for review.

3. Project Objectives

The project objectives are to examine available sales measures, compare BMW models, investigate geographic differences, explore vehicle attributes, and communicate results through interactive dashboards. The analysis should support questions such as which models or regions lead under a selected measure, how measures change across years, and how attribute categories compare. The answers depend on the definitions and completeness of the supplied dataset.

4. Scope and Boundaries

Included scope

- Dataset review and field interpretation for the 2010–2024 analysis period.
- SQL-based data overview, quality checks, KPI analysis, product/market analysis, and advanced sales analysis.
- Four Tableau dashboards plus a Home/navigation page.
- Exported dashboard images and project documentation.

Out of scope unless separately verified

- Official BMW corporate financial reporting or audited company-wide sales figures.
- Causal explanations for changes in sales, such as marketing campaigns or economic events.
- Forecasts, profitability conclusions, or customer-level insights not supported by fields in the dataset.

5. Dataset Overview

The project uses a CSV dataset described as BMW sales data for 2010–2024. The fields discussed in the project include model, year, region, color, fuel type, transmission, engine size, mileage, price in USD, and sales volume. Confirm the exact column names, units, row count, missing-value count, and dataset provenance directly from the source file before publishing final claims.

Important interpretation note

The meaning of each row determines which aggregations are valid. For example, multiplying price by sales volume to estimate revenue is appropriate only if price and sales volume have compatible meanings and the row represents the intended unit of analysis. A sum of prices alone is not necessarily revenue. Likewise, average mileage and total mileage answer different questions.

6. Tools and Their Roles

MySQL supports data inspection, grouping, filtering, aggregations, subqueries, and analytical result generation. Tableau Public supports interactive visualization, KPI presentation, filters, and dashboard navigation. Microsoft Excel can be used for quick inspection or supplementary checks. CSV is the input data format identified in the project notes.

7. Deliverables and Folder Structure

- **1_Data:** source dataset and any clearly labeled supporting data files.
- **2_SQL:** main SQL script and organized query-result folders.
- **3_Tableau:** Tableau workbook and related project files.
- **4_Dashboard_Exports:** Home image and Dashboard 1–4 image exports.
- **5_Documentation:** project documentation and methodology notes.
- **6_Reports:** final PDF reports for sharing and review.

8. Quality and Reproducibility Checklist

- Confirm dataset filename, period, field names, and row count.
- Check missing values, duplicate records, invalid ranges, and category spelling.
- Ensure each KPI uses the correct aggregation and units.
- Verify dashboard filters affect the intended worksheets.
- Re-run representative SQL queries and compare their outputs with Tableau.

- Keep a packaged Tableau workbook where possible and retain the source SQL script.

9. Limitations and Responsible Interpretation

The results are limited by the dataset’s coverage, definitions, quality, and provenance. Unless the data source is independently verified, describe the work as an analysis of the provided BMW sales dataset rather than official BMW company reporting. Report observed patterns as associations or comparisons, not causes. Do not publish numerical findings until the corresponding query output has been checked.

10. Final Outcome

The project establishes an end-to-end analytics portfolio artifact: data inspection, SQL analysis, dashboard visualization, export management, and documentation. Its professional value comes from transparent methodology, reproducible queries, accurate KPI definitions, clear visual communication, and findings that are traceable to the dataset.